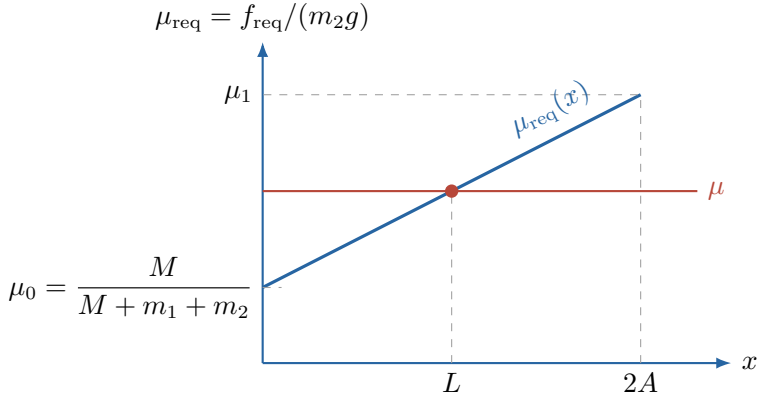
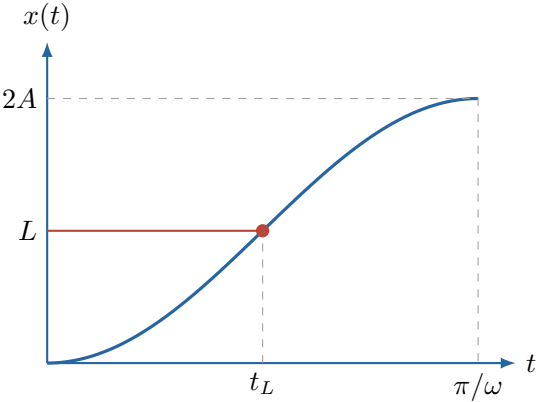


$A = Mg/k$



$x = A(1 - \cos \omega t)$